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IoT Assignment 2

Part A: Circuit Calculations

1. Circuit a (Series Configuration)

Components: 125V Supply, Resistors 20Ω, 30Ω, 50Ω.

- **Total Resistance:** 20 + 30 + 50 = 100 Ω
- **Total Current:** 125V / 100Ω = 1.25 A
- **Current through each resistor:** 1.25 A
- **Voltage Drops:**
  - 20Ω Resistor: 1.25A × 20Ω = 25V
  - 30Ω Resistor: 1.25A × 30Ω = 37.5V
  - 50Ω Resistor: 1.25A × 50Ω = 62.5V
- **Power Dissipation:**
  - 20Ω Resistor: 25V × 1.25A = 31.25W
  - 30Ω Resistor: 37.5V × 1.25A = 46.875W
  - 50Ω Resistor: 62.5V × 1.25A = 78.125W

Note: For the visual representation of this circuit, please refer to the attached schematics file.

2. Circuit b (Parallel Configuration)

Components: 125V Supply, Resistors 20Ω, 100Ω, 50Ω.

- **Total Resistance:** 1 / (1/20 + 1/100 + 1/50) = 1 / (0.05 + 0.01 + 0.02) = 1 / 0.08 = 12.5 Ω
- **Total Current:** 125V / 12.5Ω = 10 A
- **Voltage Drop through each resistor:** 125V
- **Current through each resistor:**
  - 20Ω Resistor: 125V / 20Ω = 6.25A
  - 100Ω Resistor: 125V / 100Ω = 1.25A
  - 50Ω Resistor: 125V / 50Ω = 2.5A
- **Power Dissipation:**
  - 20Ω Resistor: 125V × 6.25A = 781.25W
  - 100Ω Resistor: 125V × 1.25A = 156.25W
  - 50Ω Resistor: 125V × 2.5A = 312.5W

Note: For the visual representation of this circuit, please refer to the attached schematics file.

Part B: Component Lists

Table 1: Circuit A & B Components

Component Name	Specifications	Quantity
DC Voltage Source	125V	2
Resistor	20 Ω	2
Resistor	30 Ω	1
Resistor	50 Ω	2
Resistor	100 Ω	1

Table 2: Arduino Nano & DHT22 System

Component Name	Part/Value	Purpose
Microcontroller	Arduino Nano v3.x	Main processing unit
Humidity/Temp Sensor	DHT22 (AM2302)	Data acquisition
Pull-up Resistor	10k Ω	Signal stabilization

Note: For the Arduino/DHT22 schematic diagram, please refer to the attached schematics file.